

Great Ape Questions

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Abstract

Various lines of evidence suggest that non-human great apes are incapable of asking questions. From this, one might claim that the communicative function of question-asking is among the various features that make human communication unique. However, such uniqueness claims mystify the origins of our communicative practices, both obscuring the development of these practices along with their underlying functions.

Thus, in this paper I challenge the claim that great apes lack the ability to ask questions by identifying a novel account of question-asking compatible with their known communicative abilities. On this account, question-asking, minimally speaking, is the communicative manifestation of a species' innate capacity for curiosity. Ultimately, whether question-asking is present within great ape communicative repertoire remains an open empirical question, yet theorizing about such minimal forms offers insight towards the evolution and foundations of communicative acts.

1 A Deficiency

Call the claim that non-human great apes (or simply “great apes”) cannot ask questions “The Deficiency Hypothesis”. More precisely,

DH. The communicative capacities of great apes are deficient for question-asking.

Importantly, **DH** relates the capacities underlying the communicative repertoire of great apes to a particular *communicative act*, namely question-asking.¹ Ac-

¹See Murray and Starr (2020) for similar use of terminology and modeling

cordingly, *question-asking* is meant to pick out a class of communicative acts characterized by their typical *function* or *force*. This contrasts *questions* as linguistic entities, which, when tokened in an utterance, typically function to ask a question. The crucial point is that we can investigate the nature of communicative acts (i.e., question-asking) independent of how they are linguistically realized in human communication (i.e., via questions).²

Support for **DH** comes from a mix of both empirical observations and theoretical considerations. Here I canvass three lines of support: (i) enculturated great ape studies indicate a general lack of question-asking; (ii) question-asking is absent from typologies of great ape communication in the wild, (iii) theoretical accounts of question-asking require capacities beyond the ability of great apes.

Early attempts to teach captive great apes forms of symbolic language generally report a lack of question-asking. Either by sign language or lexigrams many great apes were trained in forms of symbolic communication in order to assess their grammatical abilities (e.g., Savage-rumbaugh et al. (1993); Terrace et al. (1979); Premack and Premack (1972)). Across this research program there seems to be a general consensus that great apes have the ability to appropriately respond to questions, but they don't ask questions themselves. For instance, Sarah, a captive chimpanzee, was taught to communicate via a variety of symbolic tokens. Contained within this language were place-holder tokens that served as an interrogative element (e.g., similar to "*who* ate the cake" versus "Trevor ate the cake"). Despite the fact that Sarah responded well by replacing the symbols appropriately, they never ask posed questions on their own (Premack and Premack (1972)). Similarly, Kanzi, a bonobo famously known to have acquired symbolic communication through third-party observation, demonstrated a remarkable ability to respond to a variety of requests, but there is no evidence of him posing any questions in return. Though much controversy shrouds these studies (e.g., Rivas (2005)), making it difficult to draw decisive conclusions, they nonetheless provide *prima facie* support for **DH**.

More recent efforts to understand great ape communication in the wild show little to no evidence of question-asking. While there is little doubt that great ape communicative systems are intentional — they are flexibly employed in pursuit of

²Pragmatics, conceived as the study of communicative acts, enjoys a certain kind of *autonomy* from linguistic theorizing (e.g., Stalnaker (2014); Armstrong (2018)).

apparent goals and are sensitive to aspects of their recipients (Liebal et al. (2004a,b); Hobaiter and Byrne (2011b,a)) — many accounts of their gestural communication, in particular, do not indicate any question-asking meaning or function. For example, Cartmill and Byrne (2010) identify gesture meanings of wild orangutans as a cluster of imperative requests, absent question-like functions: Affiliate/Play, Stop action, Look at/Take object, Share food/object, Co-locomote, and Move away. Given overlap in gestural repertoires, these meanings are likely stable across great ape species.

Despite the fact that much of the support for **DH** comes from the absence of observed question-asking, rather than direct evidence of a deficiency, there are additional theoretical considerations that independently support the hypothesis. More theoretically driven support comes from the fact that many accounts of question-asking require cognitive capacities that, at the very least, are controversially attributed to great apes. From the view of Speech Act Theory question-asking is a special form of *directive* where speakers express an attitude, a desire in particular, in expectation of an interlocutors responsive action.³ Question-asking is distinctive because, in expressing their desire, speakers request *information*. Characterized as a request/desire for information, question-asking therefore requires mental state attribution: communicators must attribute mental-states containing the relevant information to their recipients. Likewise, recent work in psycho-linguistics (Heller and Brown-Schmidt (2023); Heller and Brown-Schmidt (2025)) suggests that the comprehension of questions require the tracking of multiple perspectives, evidencing mechanisms that allow communicators to monitor the similarity/dissimilarity of their own perspective relative to representations of their interlocutor.

Thus, both older philosophical theorizing and new experimental work point to the need for mental-state attribution. While great-ape communication is certainly intentional, it appears primarily to exhibit first-order intentionality (Townsend et al. (2017); Dennett (1983)). That is, great apes produce flexible communicative acts guided by goals that are sensitive to their recipients' behavior. What remains controversial is whether these acts involve the attribution of mental states to others — this would require a level of second-order intentionality — a necessary requirement for the above accounts of question-asking.

Similar considerations have led some to theorize that great ape communica-

³Notably, Searle (1969); Bach and Harnish (1979); Vanderveken (1990)

tion serves primarily an imperative function Tomasello (2008). That is, much (if not all) great ape communication serves to coordinate typical social interactions, such as play, grooming, and copulation (Cartmill and Byrne, 2010; Tomasello and Call, 2018). However, it is not clear that many of these interactions are aptly characterized as *information-sharing*, and experimental attempts to elicit such interactions have largely produced negative results (Tomasello et al., 2005). Consequently, some interpret this as evidence that great apes lack communicative systems with declarative functions. The same reasoning extends to question-asking: if question-asking is inherently defined by its association with declarative responses, then the apparent absence of such interactions in great apes might seem to suggest that they also lack the capacity for question-asking.

2 A Curious Case

Despite the various lines of support, the truth of **DH** should strike one as surprising for several reasons. First, question-asking appears to be a fundamental feature of human communication, with questions emerging as a universal type.⁴ Furthermore, even before acquiring language, infants frequently engage in various forms of interrogative pointing (Begus and Southgate, 2012). In other words, given its ubiquitous and early-emerging role in human-communication, one should expect to find forms of question-asking to be present in other communicative systems.⁵

Additionally, great apes are deeply curious. As the proximate motivation for all inquisitive behaviors, including question-asking, curiosity is considered a basic affect (Carruthers (2018); Izard (2007)). By “basic” I mean that curiosity is widespread across the animal kingdom and often drives crucial behaviors such as play and foraging.⁶ Furthermore, not only are great apes deeply curious, but the form of their curiosity is highly sensitive to social factors (Forss and Willems (2022)). For example, chimpanzees are more curious about social interactions than about observing individual conspecifics (Lewis et al., 2025).

Lastly, despite lingering controversies about great apes abilities to attribute

⁴See König and Siemund (2007); Sadock and Zwicky (1985).

⁵Thanks to XX for emphasizing this point (p.c.)

⁶See Calhoun et al. (2014) for forms a curiosity present in round worms.

mental-states, it's clear that they have some means for tracking the epistemic status of other conspecifics via perceptual reasoning (Hare et al. (2000); Karg et al. (2015)). Moreover, these mechanisms appear to be employed in communication. For instance, Crockford et al. (2012) found that wild chimpanzee alarm calls were sensitive to the ignorance of conspecifics: individuals produced more calls when their recipients were unaware of a threat than conspecifics who were already informed.

Taken together, these considerations cast doubt on the evidence supporting **DH**. The absence of question-asking in great apes (whether in wild gesture typologies or in studies of enculturated apes) may reflect ecological or methodological factors rather than a genuine deficiency. Great apes possess the motivational profile and socio-cognitive communicative capacities that question-asking appears to require: they are curious and exhibit intentional communication that is sensitive to others' knowledge states. For these reasons, the next section develops a novel account of question-asking grounded in innate mechanisms of curiosity that may already operate within great ape communication.

3 A Minimal Account of Question-Asking

In this section, I present and motivate an account of question-asking that is compatible with the communicative capacities of great apes. The guiding idea is to theorize how question-asking could emerge in organisms with innate curiosity and how minimal forms of it might appear within their communicative systems. After presenting the account, I outline various ways it might be realized within great ape communication and explain how it is compatible with their systems, which exhibit first-order intentionality. Finally, I motivate the account by highlighting the advantages it holds over other theories of question-asking that might lead theorists to endorse **DH**.

Again, the guiding idea is how curious organisms might manifest their behaviors within communicative systems. Despite controversy surrounding precise definitions of curiosity, it's fairly uncontroversial to broadly characterize curiosity as an affective state that intrinsically motivates behaviors that result in the reduction of ignorance/uncertainty — i.e., learning behaviors.⁷ In this way, the proximate

⁷See Kidd and Hayden (2015) and Forss et al. (2024) for broad overviews of curiosity.

trigger for curiosity is some notion of ignorance or uncertainty, such as a surprising sound or an unknown object in one's environment. In response curiosity motivates behaviors, learned or hard-wired, that result in reduction of this ignorance or uncertainty, such as exploring the location of a surprising sound or approaching an unknown object.

From this perspective, question-asking can be viewed as just another sort of learning behavior issued by curiosity. Yet, as a form of communication, it manifests as a social behavior whereby organisms rely on their interactions with conspecifics to satisfy their curiosity. To make this idea more precise and to capture how such behaviors might minimally appear within a communicative system, I propose the following characterization:

Minimal Question-Asking:

Agent C asks a question via action ϕ to an interlocutor I if and only if:

1. ϕ is a characteristic signal expression of ignorance or uncertainty.
2. ϕ is selectively produced in the presence of individuals who typically act in a way to reduce C 's ignorance or uncertainty.

To see how this might be realized within communicative systems of great apes, consider an imaginary case where a group of chimpanzees exhibit variation in alarm calls that pattern with the specificity of a potential predator. For instance, these chimpanzees might have distinct calls for certain predators, like leopards or snakes, whereas another call might only signal the general or ambiguous presence of a predator. Consequently, a chimpanzee might manifest a form of Minimal Question-Asking when it produces the general alarm call in the presence of others who, in response, typically produce the more specific calls.

To paint a broader picture, consider how Minimal Question-Asking might also be realized in gestural systems of great-apes. Tomasello and Call (2018) distinguish between two types of gesture forms — *intention-movements* and *attention-getters*. On the former, certain gestures acquire their meaning through repeated use in social interactions. Over repeated occurrences, these gestures acquire meaning through their learned place among the interactions. For example, a baby chimpanzee who raises both of their arms might signal to their mother to pick them up. The gesture of arm raising gains its meaning over time as the gesture, "raising

one's arms," is reliably met with the response: "being picked up". A similar story could be told regarding Minimal Question-Asking. For instance, suppose an infant chimpanzee curiously grasps at novel or unknown objects just out of reach, but within their mother's reach. Over repeated interactions where the mother assists the infant in satisfying their curiosity, the initial grasping gestures which are characteristic of the infants curiosity may acquire a Minimal Question-Asking function.

On the other hand, attention-getters operate by directing the attention of conspecifics towards species-typical displays. More precisely, through a variety of gestural means (e.g., chest beating, leaf-clipping, ground slapping, etc...) individuals direct the attention of conspecifics towards species-typical expressions such as a playful face and bodily posture. The meanings of these gestures are therefore derived from accompanied species-typical displays. In a similar manner, Minimal Question-Asking might be realized through attention-getting gestures that direct attention toward an individual's outward expression of ignorance or uncertainty, thereby signaling the individual's curiosity.

Now, it is worth emphasizing that Minimal Question-Asking only requires first-order intentionality — i.e., it doesn't require that individuals (great-apes or others) attribute mental-states. More precisely, success in the imagined cases of question-asking above does not require the attribution of ignorance or uncertainty to the communicators. For instance, in the alarm call example, it's easy to imagine the adaptive significance of a mechanism whereby individuals produce alarm calls in the presence of conspecifics who are also alarm calling. In other words, hearers of the alarm call do not need to respond to other calls by attributing ignorance/uncertainty nor do they need to attribute an intention to reduce it to the initial callers. Likewise, the imagined cases of intention-movements and attention-getters only require that recipients have acquired some social competency (learned or hardwired) of linking behaviors characteristic of ignorance along with their desired responses. For example, in the intention-movement case, all that is required is that the mother has some means to recognize that the curious grasping of their infant is appropriately met with bringing the object within reach. The mother doesn't have to represent the infant as curious or ignorant.

Given the lack of mental-state attribution, Minimal Question-Asking holds certain advantages over accounts of question-asking that support **DH**. First, Mini-

mal Question-Asking enjoys a form of parsimony. At its core, the account treats question-asking as a communicative function individuated by its motivational basis — i.e., curiosity. This contrasts accounts, such as Speech Act Theory, where question asking is tied to particular representational capacities, thereby limiting their applicability. By contrast, this account allows for the possibility that functionally similar behaviors might emerge across curious organisms with varying representational capacities without positing discontinuous cognitive leaps.⁸

Second, by individuating question-asking through its motivational basis, the account broadens the scope of question-asking beyond declarative responses. Recall that some (e.g., Tomasello (2008); Tomasello and Call (2018)) claim that great ape communication is solely imperative in function, because it is controversial to attribute declarative uses to their communicative systems. On similar grounds, one might be tempted to preclude question-asking if it were defined as requiring declarative responses. However, reducing the function of question-asking to assertives or declaratives has a poor track record, dating back to early theorizing about the semantics of questions and question-asking.⁹ For, questions can be answered by a range of communicative functions—promises, expressives, imperatives, and more. For instance, *What did you think about the conference?* may be answered with *Exhilarating!*, and *What do you think about this mole?* with *Call your doctor*. Thus, Minimal Question-Asking does not require that an antecedent declarative function be present within a communicative system in order for question-asking to arise.

4 Minimal Questions in the Wild

Having identified and defended an account of question-asking compatible with the communicative capacities of great apes, I shift focus from whether they *can't* ask questions to whether they in fact *do*. Though, any serious answer to this question requires careful empirical analysis, I reconsider some of the initial evidence

⁸On my view, affective state, such as curious, may manifest in organisms of varying capacities for representation. For instance, it is only when affective states are accompanied with representations that they become *emotions* (Burge (2022)). Similarly, I take it that curiosity can manifest at a variety of representation levels with the highest being something like a second-order desire to know. See Carruthers (Forthcoming) for lower levels.

⁹See Vanderveken (1990) and Groenendijk and Stokhof (2011).

supporting the Deficiency Hypothesis. In particular, I allude to aspects of the methodology of studying great-ape gestures that might leave open the possibility of Minimal Question-Asking. Finally, whether or not Minimal Question-Asking is actually realized in great ape communication, the account remains theoretically useful. By minimizing representational demands, it provides a framework to explore alternative explanations for the apparent absence of question-asking in great apes and to trace the developmental pathway from this minimal form to the complex question-asking observed in human communication.

Having defended this account, we can now reconsider some of the empirical evidence cited in support of **DH**. In particular, I focus on evidence from the gestural systems of wild great apes. After all, this support for **DH** is tenuous at best: the absence of observed question-asking does not entail a deficiency nor that these behaviors don't exist. In fact, I take it that the evidence still leaves open the possibility some great ape gestures may be performed with a minimal question-asking function.

To see this, we have to get into the details of how great ape gestural meanings are individuated. In the relevant typologies (such as, Cartmill and Byrne (2010); Hobaiter and Byrne (2017); Graham et al. (2018)), the meanings or functions of particular gestures are inferred from "Apparently Satisfactory Outcomes" (ASOs). ASOs are the immediate outcomes or interactions that "satisfy" a gesturer. That is, the gesturer stops signing in response to the interaction/outcome. After identifying a satisfactory gesture instance, researchers then attribute meaning to the gesture by inferring the intention likely responsible for producing the observed ASO.

However, it's clear that this sort of methodology identifies meanings in a way too coarse-grained to identify question-asking. Focusing solely on outcomes cannot distinguish whether a gesture functions as an imperative or as a question. For instance, reconsider the case of Minimal Question-Asking between the infant chimp and their mother. What differentiates a question-asking gesture from an imperative request for, say, food, is that the gesture is proximately motivated by curiosity. Yet, from the researcher's perspective, the outcome of a food request may appear identical to curiously grasping at a food source. Thus, a methodology of individuating gesture meanings needs to be able to identify the intentions or motivations of a gestural act independent of outcome. Developing such method-

ologies requires considerable ingenuity, but until they are available, the question of whether great apes ask questions remains open.

5 Polemics

Before concluding, I consider several challenges and criticisms, some of which were set aside in earlier sections for the sake of exposition

Locating Question-Asking in Great Apes. How might one definitively establish that great apes ask questions? The arguments in the preceding section suggest that this remains very much an open empirical question. So what would the initial steps of a systematic account of question-asking in great apes look like?

One way to react to the Minimal Question-Asking Account is to think that all one needs is a way to measure curiosity as it manifests in the communicative systems of great apes. While doing so would partly settle the question of whether great apes ask questions, it would leave us with an impoverished account of question-asking. The bottom line is that we want a broader account of how question-asking, as a communicative practice, emerges within the social lives of great apes. In other words, we want an account that identifies the role of question-asking within their social interactions—the problems it solves and the ways in which it is employed. Only with such a fuller framework can one begin to illuminate aspects of question-asking as they emerge in human communication.

Indeed, a major upshot of the foregoing view is a shift in focus away from theorizing about question-asking in great apes as a function of their meta-representational capacities, and toward the ways great apes actually employ their curiosity: how they flexibly deploy adaptive behaviors to overcome ignorance or uncertainty about their environment. By focusing on patterns in their inquisitive behaviors, we are able to profile the cognitive styles of great apes as they pertain to differences and similarities in curiosity across species of great apes (including humans), other primates, and other highly communicative species. For instance, chimpanzees, like humans, tend to be more risk-averse when uncertainty stems from social factors, as opposed to non-social risk Haux et al. (2021). Similarly, Forss and Willems (2022) found that more social great ape species, like bonobos, employ fewer curious behaviors when alone than more solitary great apes, like orangutans. Furthermore, Carruthers and Williams (2019) interpret differences in information-seeking found

in rhesus and capuchin monkeys (Beran and Smith, 2011) as indicating that rhesus monkeys exhibit a more cautious and reflexive style. By profiling the curious nature of great apes, we are able to better situate where, within their communicative interactions, question-asking might emerge and the roles it plays within these interactions.

Consequently, a systematic approach to question-asking benefits, in addition to attending to the factors that shape curiosity, from theorizing about the *ways* curiosity promotes adaptive solutions *independent* of social or communicative interactions. Though I take it that great apes' curious capacities are phylogenetically prior to question-asking, even if the reverse were true, we have strong reason to think that the two are intimately linked. To this end, a variety of experimental paradigms, especially those falling under the label of *information-seeking*, support the claim that great apes possess inquisitive capacities comparable, in relevant respects, to the ways we ask questions in language.

That is, great apes employ information-seeking behaviors in ways that reflect our ability to answer questions such as where, what, how/why, and who. Recently, Rosati et al. (2024) validated previous work (a.o., (Call and Carpenter, 2001; Beran et al., 2013)) demonstrating that non-human primates engage in information-seeking behaviors to overcome ignorance about an object's location.¹⁰ That is, chimpanzees will adjust their perspectives to gain information about the location—the where—of a reward. Additionally, Rosati et al. (2024) showed that chimpanzees will employ these behaviors not only to gain information about location, but also about the identity of the reward—the what. In a similar paradigm, Bohn et al. (2017) demonstrated that great apes engage in similar behaviors to gain information about the properties of tools (e.g., the length of a rake) in order to use them later to obtain a reward, suggesting a capacity to understand how the tool will be used. Though considerable conceptual challenges remain in developing experimental paradigms that capture the explanatory capacities of non-human primates ((Andrews, 2012; Völter et al., 2020)), Wang and Hayden (2018) found that

¹⁰Despite these advancements, much of the work under the information-seeking paradigm is limited. Given the highly experimental settings, these paradigms fail to illuminate the ecological validity of these behaviors. In other words, it is difficult to determine how great apes typically employ these behaviors to overcome ignorance in their natural environment. Second, as many authors note, these experiments do not control for curiosity: whether the relevant great apes value the information gained for its own sake, as opposed to valuing it through some form of indirect reinforcement.

macaques are curious about counterfactual outcomes even at the cost of material reward.

Lastly, recent evidence suggests that great apes are curious about other social agents—i.e., who. As mentioned previously, Lewis et al. (2025) found that chimpanzees will forgo instrumental rewards (e.g., food) in order to gain information about social interactions. Such a capacity is adaptive when individuals need to acquire information to navigate their social worlds. Furthermore, Carvajal and Krupenye (2025) found that a bonobo, Kanzi, can concurrently track the location of agents—who is where—by integrating information from sight and hearing. It should be noted that while this evidences great apes’ capacities for curiosity in social interactions, these are not social interactions in which other agents help satisfy one’s curiosity, as described by the Minimal Question-Asking Account. Rather, what is required for the foregoing account is that the behaviors employed aim to strategically gain information with the help of another agent.

To this end, much of what we know about the curious profile of great apes suggests that we are most likely to find question-asking present in food-sharing interactions. This is due to several reasons. First, food sharing interactions are often communicative. As previously suggested in Cartmill and Byrne (2010), great-apes employ gestures to coordinate food-sharing interactions. Second, food sharing, in particular between infant and caretaker, are situated among the most cooperative types of social interactions. In such interactions, there are many adaptive reasons why others might not take advantage of ignorant agents. Thus, they provide contexts in which inquisitive behavior might emerge.

Last, and most importantly, food sharing interactions between infants and caretakers are facilitated by curiosity. In other words, they are not guided by the instrumental value tied to food. Jaeggi et al. (2008) found that both infant orangutan solicitation rates and varieties of food were not explained by nutritional value. For instance, infant orangutans did not solicit higher-nutritious foods more frequently than lower ones. Instead, Jaeggi et al. (2008) found that both the socialization rates and varieties of food were better predicted by the fact that these solicitations aimed at learning about the food. Infant orangutans solicited mostly for foods that were difficult to process, and their solicitation rates dropped as their individual ability to forage increased. Furthermore, in a follow up study, Mikeliban et al. (2021) not only replicated these findings, but also illuminated more of role of

the food-sharers. They found that mothers' social tolerance for food sharing also adjusted with the complexity of the food, suggesting that these interactions were focused on learning. In the end, our best understanding of the curious capacities of great apes suggests that they manifest some form of question-asking. However, definitive conclusions should be foregone until further research establishes (i) the precise role of curiosity in food-sharing interactions, and (ii) how these interactions are linked to the gestures used to solicit food.

Why Curiosity? In previous sections, it was taken for granted that curiosity is the main motivating mechanism behind inquisitive behaviors, such as question-asking. Though accepted on intuitive grounds, the discussion thus far opens up space to consider other potential mental state/mechanisms: why not *Ignorance* or *Uncertainty*? After all, the Minimal Account makes essential appeal to expressions of ignorance. Furthermore, why not mechanisms for learning and information-seeking more broadly? That is, what is added by the further constraint of curiosity that the learning must be *intrinsically motivated*?

So, why not *Ignorance*? After all, the Minimal Question-Asking account posits expressions of Ignorance as necessary conditions on minimal question-asking.¹¹ Thus, what's wrong with thinking that states of ignorance are all that is required to manifest question-asking in communication?

The issue is that ignorance, qua mental state, on its own does not motivate inquisitive behaviors. Considering the role of ignorance in relation to curiosity, it functions primarily as a trigger. A dog, for instance, might approach a novel toy out of ignorance of its properties. However, ignorance does not always lead to curiosity and subsequent information-seeking behaviors. There are many facts I am ignorant of—e.g., the number of hairs on my head or the number of leaves on the tree outside my window—where the most adaptive response is *not* to seek out answers. Rather, it's the function of curiosity, as an affective mechanism, to evaluate instances of ignorance as worthy of motivating inquisitive behaviors.¹² In this sense, curiosity is crucial to question-asking because, unlike ignorance, it is

¹¹The Minimal Question-Asking account also appeals to *Uncertainty*. Though, there are theoretical reasons to distinguish Ignorance and Uncertainty, likewise for distinctions within Ignorance and Uncertainty, I treat both as one and the same. For, the general argumentative point applies equally.

¹²Thus, the Minimal Account appeals to expressions of ignorance, which are ultimately caused by curious mechanisms that evaluate whether ignorance is worth resolving.

curiosity that ultimately motivates inquisitive behaviors.

Though ignorance is insufficient of question-asking, some might still deny that curiosity individuates question-asking, in favor of more general learning mechanisms. The point being that question-asking need *not* be *intrinsically motivated*. In support of this view, is the mundane fact that we often ask-questions in language to inquire into instrumental aims. In response to this line of thought, my response is two-fold.

First, the intrinsic motivation from curiosity delineates the communicative function of question-asking from other acts. The fact that curiosity is intrinsically motivated clarifies the *characteristic* aim/function of question-asking: namely, to acquire information. This point is especially salient when delineating question-asking from other directive acts. While, other communication functions may result to have the force/function those only do so instrumentally or indirectly. For instance, one might have the command of *tell me where the cookies are!* has the effect or force of acquiring information. Yet, the point is that the information that the command aims at is value instrumentally, for the purposes of finding and, presumably eating cookies! Thus, the fact that curiosity intrinsically motivates information-seeking is what individuates it from other communicative acts and their functions.

Second, that said, question-asking need not always be intrinsically motivated. In presenting the Minimal Question-Asking account, the aim is to identify the *etiological function of question-asking*—that is, how it arose as a distinct form of communication and why it continues to persist. As in standard theories of speech acts, there is no one-to-one mapping between *content* and *force*—in this case, between the mental states underlying communicative acts and their effects. In the same way that a phone camera can be used as a flashlight, the function of a behavior can be adapted for alternative ends. The crucial point is that curiosity explains the characteristic use of question-asking. It is a further claim that, in human language, question-asking has adapted to serve instrumental ends — a trajectory that the present framework seeks to illuminate.

Second, that said, question-asking need not always be intrinsically motivated. The Minimal Question-Asking account seeks to explain the *etiological function of question-asking*—how it arose as a distinct form of communication and why it persists. As in standard speech-act theory, there is no one-to-one mapping between

content and *force*; similarly, the mental states underlying a communicative act do not rigidly determine its effects. Just as a phone camera can be repurposed as a flashlight, the function of a behavior can be adapted to serve alternative ends. In this light, curiosity provides the explanation for the typical or characteristic use of question-asking, while instrumental or goal-directed uses can emerge as functional adaptations. The present framework thus traces the trajectory from minimal, curiosity-driven question-asking to the complex, often instrumental forms seen in human linguistic practice.

6 Conclusion

Nonetheless, even if it turns out that great apes do not ask questions, the Minimal Question-Asking account remains theoretically fruitful. Supposing that no form of minimal question-asking is found in great ape communication, the account still provides a framework for understanding why question-asking may be absent. For example, because Minimal Question-Asking is compatible with great apes' communicative capacities, one might look to explanations alternative to their communicative abilities that prevent its emergence, such as differences in motivation and curiosity or ecological constraints.

Similarly, as a first-order account of intentional question-asking the account also provides a fruitful starting point for theorizing about the trajectories of question-asking as the representational capacities of communicators vary. In other words, how do we get from Minimal Question-Asking to question-asking as it appears in human communication? And how might differences in representational capacities shape those trajectories? Ultimately, I hope that this account illuminates potential pathways for theorizing about question-asking, offering a framework for understanding its emergence and evolution in both human and non-human great apes.

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